[Total No. of Question -04] [Total No. of Pages - 01]

G H Raisoni International Skill Tech University, Pune

S.Y. B. Tech (CE/IT/Cy. Sec)
(Sem-IV)
CAE -II- SUMMER 2026 (2025 Pattern)
Open Elective-I: Web Technology (25BTOECE01A)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) Solve Q.1 or 2 and Q3 or 4. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1. (a)	Design and implement a Node.js HTTP Server to serve HTTP responses	[5]	3	3
b)	Describe a client server request response cycle in Node.js with the help of suitable example	[5]	3	2
	OR			
Q 2. (a)	Describe the role of Express.js framework and middleware in node.js applications	[5]	3	2
(b)	Write code and give steps to Develop an Express.js application that handles both GET and POST requests. Write code for at least two routes	[5]	3	3
Q 3. (a)	Differentiate between SQL and NoSQL Databases. Explain the document-based structure of MongoDB with the help of an example showing a document and its corresponding collections	[5]	4	2
(b)	Give steps in connecting a MongoDB database to a node.js Application. Explain the role of MongoDB Driver or Mongoose ODM in managing the connection	[5]	4	2
	OR			
Q. 4 (a)	Explain four basic CRUD operations of any database. Provide the insert, read, update and delete shell commands of MongoDB	[5]	4	2
(b)	Describe the structure of MongoDB BSON Document format. Give suitable example	[5]	4	2